

Module 7 Homework

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1. Generate a map of your home location

```
#getting ohio map
map.county <- map_data('county')
ohio.county <- subset(map.county, region=="ohio")

#hometown coordinates
wapak_df <- data.frame(long=-84.19408, lat=40.56894)

#US map
states_map <- map_data("state")

#plot of my hometown
ggplot() +
  geom_polygon(data = states_map, aes(x=long,y=lat,group=group), fill="white", colour=
    "black") +
  geom_point(data = wapak_df, aes(x = long, y = lat)) +
  annotate(geom="text",x=-84.19408,y=40.56894, label="Wapakoneta",adj=-0.05, color="red")+
  theme_nothing()
```



2. Generate a choropleth map of Ohio and shade counties by their case rate per thousand

```
#get corona cases from ohio.gov
ohioDF <- read_csv("https://coronavirus.ohio.gov/static/COVIDSummaryData.csv")
#get ohio population from census.gov source = https://www.census.gov/data/datasets/time-
#series/demo/popest/2010s-counties-total.html#par_textimage_739801612
ohioPop <- read_excel("co-est2019-annres-39.xlsx")

#get the date
Today <- Sys.Date()

#filter out row of totals and mutate variables
ohioDF <- ohioDF %>%
  filter(Sex != "Total") %>%
  mutate(OnsetDate = mdy('Onset Date')) %>%
  filter(Today - OnsetDate <= 30) %>%
  group_by(County) %>%
  summarize(ncases = sum('Case Count'))

#filter out unneeded rows and columns before combining tables
ohioPop <- ohioPop[c(-1, -2, -3, -4, -93, -94, -95, -96, -97),2]
```

```

#combine tables and changing variables
#rename county to region to map it with ohio.county
#https://www.sharpsightlabs.com/blog/rename-columns-in-r/
combinedOhio <- cbind(ohioDF, ohioPop) %>%
  mutate(countyPop = as.double(...2)) %>%
  select(!...2) %>%
  mutate(rate = (ncases/countyPop) * 1000) %>%
  mutate(County = str_to_lower(County)) %>%
  rename(subregion = County)

#combine all data sets into one
rateDB <- merge(ohio.county, combinedOhio, by.x = "subregion", by.y = "subregion")

#map
ggplot(data = rateDB, aes(x=long,y=lat,group = group, fill=rate)) +
  geom_polygon(color = "black") +
  scale_fill_gradient2(low="#559999",mid="grey90",high="#BB650B",
    midpoint=median(rateDB$rate)) +
  labs(title = "Ohio COVID-19 Infection Rates per 1000 Residents over the last 30 Days",
    subtitle = paste("Updated: ", Today)) +
  theme_map() +
  theme(legend.position = "right")

```

Ohio COVID-19 Infection Rates per 1000 Residents over the last 30 Days

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